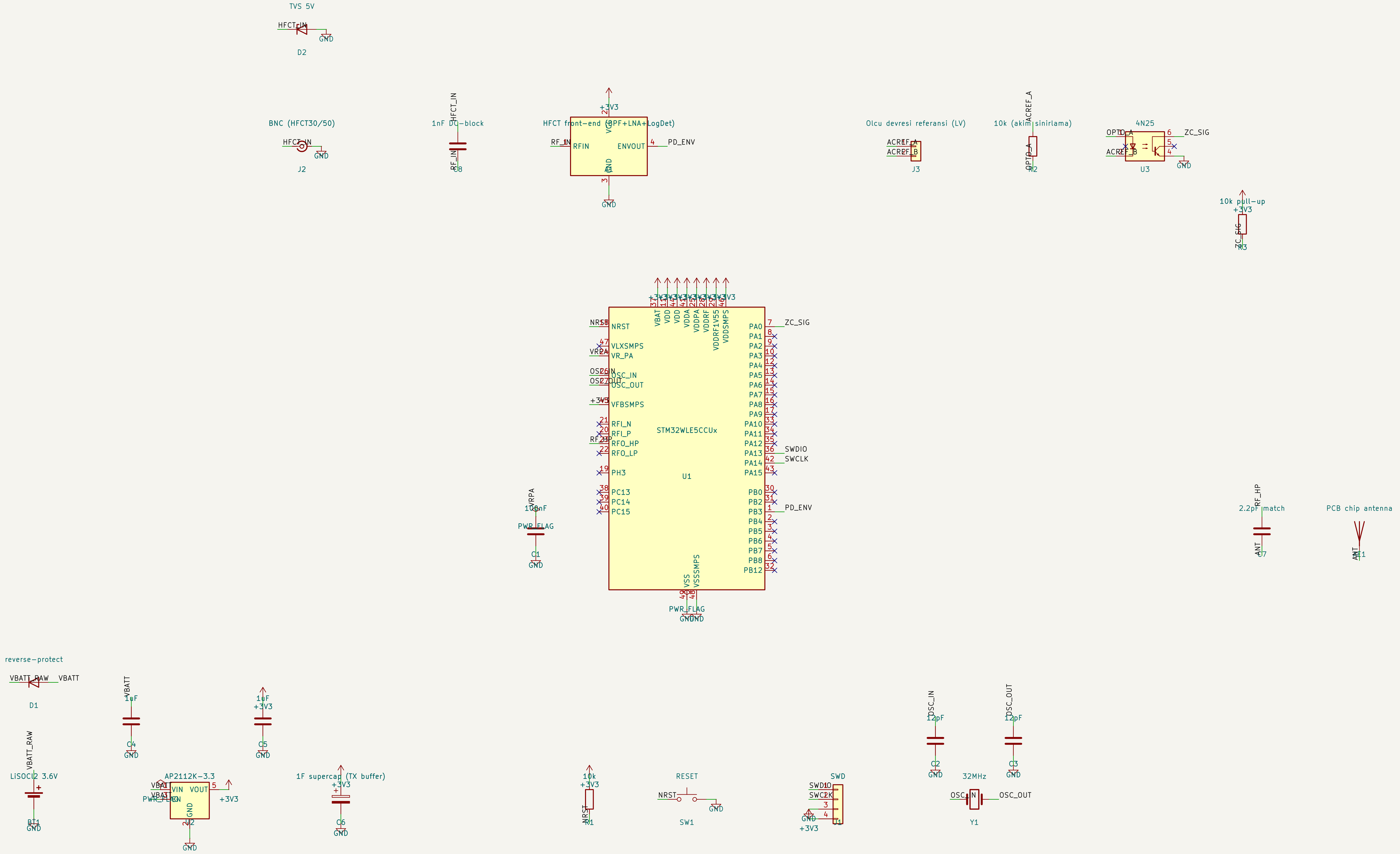
Basitlestirme: RF_Frontend kutusu bandpass filtre +
LNA + log–amp zarf dedektoru zincirini temsil eder
(gercek IC secimi PCB asamasinda, bkz. docs/03).

Faz referansi panonun Olcu Devresi (gG, >=2A) ucundan
izoleli (4N25) sekilde alinir – sifir gecis zaman damgasi icin.


TVS 5V
HFCT_IN
GND
D2
BNC (HFCT30/50)
HFCT_IN
GND
J2
1nF DC–block
HFCT_IN
RF_IN
C8
HFCT front–end (BPF+LNA+LogDet)
+3V3
VCC
RF_IN
RFIN
ENVOUT
PD_ENV
GND
A1
GND
Olcu devresi referansi (LV)
ACREF_A
ACREF_B
J3
10k (akim sinirlama)
ACREF_A
OPTO_A
R2
4N25
OPTO_A
ACREF_B
ZC_SIG
GND
U3
10k pull–up
+3V3
ZC_SIG
R3
+3V3
VBAT
VDD
VDD
VDDA
VDDPA
VDDRF
VDDRF1V55
VDDSMPS
NRST
VLXSMPS
VRPA
VR_PA
OSC_IN
OSC_OUT
+3V3
VFBSMPS
RFI_N
RFI_P
RF_HP
RFO_HP
RFO_LP
PH3
PC13
PC14
PC15
STM32WLE5CCUx
U1
PA0
ZC_SIG
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
SWDIO
PA14
SWCLK
PA15
PB0
PB2
PB3
PD_ENV
PB4
PB5
PB6
PB7
PB8
PB12
VSS
VSSSMPS
PWR_FLAG
GND
GND
VRPA
100nF
PWR_FLAG
C1
GND
2.2pF match
RF_HP
ANT
C7
PCB chip antenna
ANT
AE1
reverse–protect
VBATT_RAW
VBATT
D1
VBATT_RAW
LiSOCl2 3.6V
BT1
GND
VBATT
1uF
C4
GND
1uF
+3V3
C5
GND
AP2112K–3.3
VBATT
VBATT
PWR_FLAG
VIN
EN
VOUT
GND
+3V3
U2
GND
1F supercap (TX buffer)
+3V3
C6
GND
10k
+3V3
NRST
R1
RESET
NRST
GND
SW1
SWD
SWDIO
SWCLK
GND
+3V3
J1
OSC_IN
12pF
C2
GND
OSC_OUT
12pF
C3
GND
32MHz
OSC_IN
OSC_OUT
Y1

PANO GOZ – IR/PD dugumu (Kismi Desarj / Ark)
Ana giris kablo riseri icin (1 adet/pano)

PANO GOZ – K1
Sheet: /
File: ir–pd–node.kicad_sch
Title: PANO GOZ – IR–PD Dugumu (Kismi Desarj/Ark)
Size: A2
Date: 2026–09–13
Rev: A
KiCad E.D.A. 10.0.6
Id: 1/1